

# ctrlX OS

Version 4.6.0

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DC-AE/PAX (TaDo/PiaSt)

# Table of contents

|          |                                                                                                                                     |          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>About this documentation</b>                                                                                                     | <b>9</b> |
| <b>2</b> | <b>Important information</b>                                                                                                        | <b>9</b> |
| 2.1      | Information about our products .....                                                                                                | 9        |
| 2.2      | Information on the image update .....                                                                                               | 9        |
| 2.3      | LED display after an image update on a ctrlX DRIVEplus device can be incorrect .....                                                | 9        |
| <b>3</b> | <b>Version 4.6.0</b>                                                                                                                | <b>9</b> |
| 3.1      | Software .....                                                                                                                      | 9        |
| 3.1.1    | Components .....                                                                                                                    | 10       |
| 3.1.2    | Compatibility with drive-integrated control .....                                                                                   | 10       |
| 3.2      | New functions .....                                                                                                                 | 11       |
| 3.2.1    | Automation Core .....                                                                                                               | 11       |
|          | Feature 1114792: Pending information messages in alarms .....                                                                       | 11       |
|          | Feature 1114982: Changes in the Data Layer ....                                                                                     | 12       |
|          | Feature 1131434: Linux Kernel Update .....                                                                                          | 12       |
| 3.2.2    | Device Admin .....                                                                                                                  | 12       |
|          | Feature 282405: DHCP server support in ctrlX OS                                                                                     | 12       |
|          | Feature 1041322: "Apps" and "Licenses" were merged .....                                                                            | 12       |
|          | Feature 1051992: Support for embedding third-party applications in the ctrlX user interface via iFrame .....                        | 13       |
|          | Feature 1058176: Creating a user account during initial commissioning .....                                                         | 13       |
|          | Feature 1155681: Public keys can be copied to the clipboard .....                                                                   | 13       |
| 3.2.3    | Setup .....                                                                                                                         | 13       |
|          | Feature 1112318: Warning of downgrade .....                                                                                         | 13       |
|          | Feature 1112320: Support for additional settings .                                                                                  | 13       |
| 3.2.4    | Solutions .....                                                                                                                     | 14       |
|          | Feature 995994: Extended Git functionality .....                                                                                    | 14       |
| 3.3      | Resolved defects and modifications .....                                                                                            | 14       |
| 3.3.1    | Automation Core .....                                                                                                               | 14       |
|          | Bug 1091116: Bugfix for sporadic startup problem due to unstable eMMC enumeration .....                                             | 14       |
|          | Bug 1098779: System message "Cannot find unit for notify message of PID ****, ignoring." in the log-book .....                      | 14       |
|          | Bug 1100451: System image update from version 2.6 and older to version 4.4 via the ctrlX OS web interface results in an error ..... | 14       |
|          | Bug 1116120: Backup/restore may fail after 30 days operating time of the device .....                                               | 15       |
|          | Bug 1126017: Frequent error 081F0042/0C7B4285 when switching from OPERATING to SERVICE ....                                         | 15       |
|          | Bug 1139268: Extended time synchronization (e.g. PTP) resulted in jitter in real-time processes ....                                | 15       |

|       |                                                                                                                                           |    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|----|
| 3.3.2 | Device Admin                                                                                                                              | 15 |
|       | Bug 1076361: Time and time zone are not applied                                                                                           | 15 |
|       | Bug 1116146: ctrlX OS sporadically takes longer to boot on ctrlX CORE                                                                     | 15 |
|       | Bug 1124644: Powercontroller firmware is not updated                                                                                      | 15 |
|       | Bug 1126501: After updating all system apps on devices with image 2.6, "Upgrade ctrlX OS" (remodeling) cannot be executed                 | 16 |
|       | Bug 1129085: Image update does not work reliably on ctrlX CORE X5                                                                         | 16 |
|       | Bug 1133074: Memory leak in UPNP service                                                                                                  | 16 |
|       | Bug 1134469: Logbook missing from system report                                                                                           | 16 |
| 3.3.3 | Solutions                                                                                                                                 | 16 |
|       | Bug 1122488: Incomplete backup of Git settings                                                                                            | 16 |
| 3.4   | Notes on use and known restrictions                                                                                                       | 16 |
| 3.4.1 | Updating the apps on the ctrlX CORE Virtual                                                                                               | 16 |
| 3.4.2 | Updating the apps on ctrlX CORE                                                                                                           | 17 |
| 3.4.3 | Automation Core                                                                                                                           | 18 |
|       | Bug 589936: X3: Retain memory > 56 kB is sporadically inconsistent after switching off/on with drive-integrated control and ctrlX CORE X3 | 18 |
|       | Bug 711581: Device does not start with USB flash drive inserted on XF01C                                                                  | 19 |
|       | Bug 853922: ctrlX CORE X5: Device does not start sporadically if an external data carrier is plugged into port XF01A, XF01B or CF01       | 19 |
|       | Bug 867074: Packet loss during redundant communication with HSR                                                                           | 19 |
|       | Bug 1014687: ctrlX CORE X5/X7: First boot after image update (hardware reset) takes longer                                                | 20 |
|       | Bug 1179239: Interrupting real-time tasks at full core load                                                                               | 20 |
| 3.4.4 | Device Admin                                                                                                                              | 20 |
|       | Bug 899934: Uninstallation takes a very long time                                                                                         | 20 |
|       | Bug 1021621: Hardware support cannot be updated on a control with Secure Boot (ctrlX CORE X5/X7) from the store                           | 20 |
|       | Bug 1111609: Updating all apps on a device with image version 2.6.10 aborts sporadically after the core app                               | 20 |
|       | Bug 1188123: Resetting the device data fails                                                                                              | 20 |
|       | Bug 1189755: Uninstall button for the system app "core24" after remodeling                                                                | 21 |
|       | Bug 1192160: Creating a system report seems to freeze                                                                                     | 21 |
| 3.4.5 | Setup                                                                                                                                     | 21 |
|       | Bug 1192174: Restoring a backup on a control of a different type fails                                                                    | 21 |

|          |                                                                                       |           |
|----------|---------------------------------------------------------------------------------------|-----------|
| <b>4</b> | <b>Version 4.4.1</b>                                                                  | <b>21</b> |
| 4.1      | Software .....                                                                        | 21        |
| 4.1.1    | Components .....                                                                      | 22        |
| 4.1.2    | Compatibility with drive-integrated controls .....                                    | 23        |
| 4.2      | New functions .....                                                                   | 23        |
| 4.2.1    | Automation Core .....                                                                 | 23        |
|          | Feature 740428: Improved handling of Motion errors .....                              | 23        |
|          | Feature 870976: Functional extensions in the log-book web interface .....             | 23        |
|          | Feature 938847: Updating the Celix Framework ..                                       | 24        |
|          | Feature 984227: Updating the "core" system app ..                                     | 24        |
|          | Feature 995017: Various functional adjustments to the Data Layer web interface .....  | 24        |
|          | Feature 996130: Changes in the Data Layer .....                                       | 24        |
|          | Feature 1045881: Data Layer improvements .....                                        | 25        |
|          | Feature 1047945: Flexible core division .....                                         | 26        |
|          | Feature 1048023: Linux Kernel updated to version 6.12.40 .....                        | 26        |
|          | Feature 1059916: Functional extensions in the Data Layer web interface .....          | 26        |
| 4.2.2    | Device Admin .....                                                                    | 27        |
|          | Feature 548599: Balanced energy mode for ctrlX OS (Balanced Mode) .....               | 27        |
|          | Feature 836887: Device Admin supports CAN interfaces .....                            | 27        |
|          | Feature 914768: Mounting and unmounting storage media using Data Layer commands ..... | 27        |
|          | Feature 948328: Simple Network Management Protocol can be used in ctrlX OS .....      | 27        |
|          | Feature 986338: Notification for differing time ...                                   | 27        |
|          | Feature 986818: Supporting VLAN via DAN network interfaces .....                      | 27        |
|          | Feature 996303: System report .....                                                   | 27        |
|          | Feature 1006016: Renaming "Firmware update" to "Flash system image" .....             | 27        |
|          | Feature 1032629: Supporting Single Sign-on (SSO) .....                                | 27        |
|          | Feature 1040678: Improved display of the login banner for ctrlX OS .....              | 28        |
|          | Feature 1054708: 10-day trial period available for selected apps .....                | 28        |
|          | Feature 1060885: Displaying the ctrlX OS version in "System information" .....        | 28        |
|          | Feature 1061268: Display of remaining time in trial mode .....                        | 28        |
|          | Feature 1071645: Display CPU information in "System information" .....                | 28        |

|       |                                                                                                            |    |
|-------|------------------------------------------------------------------------------------------------------------|----|
| 4.2.3 | Setup                                                                                                      | 28 |
|       | Feature 981999: Deleting backup files after restoring                                                      | 28 |
|       | Feature 1048209: Supporting additional settings                                                            | 28 |
| 4.2.4 | Solutions                                                                                                  | 28 |
|       | Feature 995857: Configurations can be added to version control with Git                                    | 28 |
|       | Feature 995988: Collaborative engineering with Git                                                         | 29 |
| 4.3   | Resolved defects and modifications                                                                         | 29 |
| 4.3.1 | Automation Core                                                                                            | 29 |
|       | Bug 1010221: Error when creating the callable during start-up                                              | 29 |
|       | Bug 1117971: Error when automatically creating callables                                                   | 29 |
| 4.3.2 | Device Admin                                                                                               | 29 |
|       | Bug 940111: SD card can sporadically not be read in Windows PC after firmware update                       | 29 |
|       | Bug 994012: "Upgrade ctrlX OS" (remodeling) fails sporadically                                             | 30 |
|       | Bug 995534: A USB flash drive cannot be mounted again if it is removed without clicking on "remove safely" | 30 |
|       | Bug 1043154: When formatting a partition, the name of the storage medium must be specified                 | 30 |
|       | Bug 1049110: Snapd app update fails for devices without serial assertion                                   | 30 |
|       | Bug 1064879: Alarm "Failed to set CPU configuration" is displayed                                          | 30 |
|       | Bug 1079925: "Upgrading ctrlX OS" (remodeling) not possible with snapd 2.68.5.1                            | 31 |
|       | Bug 1081521: The system operating time is not updated in the Data Layer                                    | 31 |
|       | Bug 1126921: Error when restoring a subscription (certificate management)                                  | 31 |
| 4.3.3 | Setup                                                                                                      | 31 |
|       | Bug 1015711: Resetting the default password is not possible                                                | 31 |
|       | Bug 1035162: Unauthorized access to sensitive data via security mechanisms                                 | 31 |
|       | Bug 1035172: Unauthenticated access to API endpoint                                                        | 31 |
|       | Bug 1041006: Backup/restore does not support NFS and SMB mount points                                      | 32 |
|       | Bug 1053299: Unencrypted backup files                                                                      | 32 |
|       | Bug 1062953: Certain snapd versions cannot be restored                                                     | 32 |
|       | Bug 1068029: Error when restoring PKIs                                                                     | 32 |
|       | Bug 1071563: User-defined password policy is not deleted when restoring a backup                           | 32 |
|       | Bug 1076429: User is not removed from group                                                                | 32 |

|       |                                                       |                                                                                                                                         |    |
|-------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----|
|       |                                                       | Bug 1077634: Server-based licensing is not enabled when restoring a backup . . . . .                                                    | 32 |
|       |                                                       | Bug 1077648: Warning "Json schema issue: ..." when restoring a backup . . . . .                                                         | 33 |
|       |                                                       | Bug 1079332: License server is not enabled when restoring a backup . . . . .                                                            | 33 |
|       |                                                       | Bug 1113522: Certificate and PKI cannot be deleted together . . . . .                                                                   | 33 |
| 4.3.4 | Solutions . . . . .                                   |                                                                                                                                         | 33 |
|       |                                                       | Bug 970580: Authorization "Manage configurations" is insufficient for loading and saving . . . . .                                      | 33 |
|       |                                                       | Bug 1070473: Uploading archives with read-only directories fails . . . . .                                                              | 33 |
| 4.4   | Notes on use and known restrictions . . . . .         |                                                                                                                                         | 33 |
| 4.4.1 | Updating the apps on the ctrlX CORE Virtual . . . . . |                                                                                                                                         | 33 |
| 4.4.2 | Updating the apps on ctrlX CORE . . . . .             |                                                                                                                                         | 34 |
| 4.4.3 | Automation Core . . . . .                             |                                                                                                                                         | 35 |
|       |                                                       | Bug 1051857: X3: Pending error after image update . . . . .                                                                             | 35 |
|       |                                                       | Bug 1098779: System message "Cannot find unit for notify message of PID ****, ignoring." in the log-book . . . . .                      | 36 |
|       |                                                       | Bug 1100451: System image update from version 2.6 and older to version 4.4 via the ctrlX OS web interface results in an error . . . . . | 36 |
|       |                                                       | Bug 1111432: ctrlX CORE X5/X7: Warning 080E082A/0C64100D "App connection restored" after image update . . . . .                         | 36 |
|       |                                                       | Bug 1120682: ctrlX CORE X7 startup sporadically delayed by 140 seconds . . . . .                                                        | 36 |
|       |                                                       | Bug 1122688: Increased runtime of scheduler tasks . . . . .                                                                             | 36 |
|       |                                                       | Bug 1126017: Frequent error 081F0042/0C7B4285 when switching from OPERATING to SERVICE . . . .                                          | 36 |
|       |                                                       | Bug 1131271: Increasing "deviation" time measurement for scheduler tasks . . . . .                                                      | 36 |
| 4.4.4 | Device Admin . . . . .                                |                                                                                                                                         | 37 |
|       |                                                       | Bug 855538: "Reset device data" can fail sporadically . . . . .                                                                         | 37 |
|       |                                                       | Bug 1021621: Hardware support cannot be updated on a control with Secure Boot (ctrlX CORE X5/X7) from the store . . . . .               | 37 |
|       |                                                       | Bug 1076361: Time and time zone are not applied . . . . .                                                                               | 37 |
|       |                                                       | Bug 1111609: Updating all apps on a device with image version 2.6.10 is canceled after the app Core22 . . . . .                         | 37 |
|       |                                                       | Bug 1116146: ctrlX OS sporadically takes longer to boot on ctrlX CORE . . . . .                                                         | 37 |

|          |                            |                                                                                                                                     |           |
|----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          |                            | Bug 1126501: After updating all system apps on devices with image 2.6, "Upgrade ctrlX OS" (remodeling) cannot be executed . . . . . | 37        |
|          |                            | Bug 1129085: Image update does not work reliably on ctrlX CORE X5 . . . . .                                                         | 38        |
| 4.4.5    | Setup . . . . .            |                                                                                                                                     | 38        |
|          |                            | Bug 1116120: Backup/restore may fail after 30 days operating time of the device . . . . .                                           | 38        |
| 4.4.6    | Solutions . . . . .        |                                                                                                                                     | 38        |
|          |                            | Bug 1117106: Files with non-ASCII characters in the name are restored incorrectly when uploading . . .                              | 38        |
|          |                            | Bug 1122488: Incomplete backup of Git settings .                                                                                    | 38        |
| <b>5</b> | <b>Service and support</b> |                                                                                                                                     | <b>39</b> |
| <b>6</b> | <b>Index</b>               |                                                                                                                                     | <b>41</b> |

# 1 About this documentation

## Editions of this documentation

| Edition | Release date | Note                   |
|---------|--------------|------------------------|
| 01      | 2025-11      | ctrlX OS Version 4.4.1 |
| 02      | 2026-03      | ctrlX OS Version 4.6.0 |

## Further information

- See [↗ ctrlX OS, Application Manual](#)
- See [↗ ctrlX DRIVE, Firmware Release Notes AXS-V-06](#)

# 2 Important information

## 2.1 Information about our products

For the latest information on our products, go to [↗ https://www.boschrexroth.com](https://www.boschrexroth.com).

For current security advisories, please refer to [↗ https://psirt.bosch.com/security-advisories/](https://psirt.bosch.com/security-advisories/)

## 2.2 Information on the image update

The system images can be downloaded as zip archive from the Rexroth Collaboration Room.

### NOTICE

As system images are device-specific, select the zip archive compatible with your device.

When using a system image, incompatible with your device, the device can sporadically not be used anymore, i.e. the device does not boot anymore.

In this case, please contact Bosch Rexroth support.

Further information on performing the image update can be found in the documentation ctrlX OS Operating System for ctrlX CORE Control Devices", Application Manual in the chapter "[↗ Hardware reset for ctrlX CORE and ctrlX DRIVE Plus with integrated ctrlX CORE](#)".

## 2.3 LED display after an image update on a ctrlX DRIVEplus device can be incorrect



During an image update, the successful update might not be displayed correctly on a ctrlX DRIVEplus device.

For example that the LED is permanently blue or flashing red after a few seconds. At this point in time, the update is still running.

Thus, wait for 10 minutes after the start of the update.

After the waiting time, the device can be switched off, the micro SD card can be removed and the device can be switched on again.

Go to the log file on the SD card to check whether the update was successful.

# 3 Version 4.6.0

## 3.1 Software



This release is part of ctrlX CORE 4.6.0.

### 3.1.1 Components

Table 1: Included components

| Short name             | Software                                                  | Version          | Additional notes                                                                                                                                                              |
|------------------------|-----------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ctrlX Apps</b>      |                                                           |                  |                                                                                                                                                                               |
| XCR-4.6.0              | arch01.firstboot                                          | 4.6.0            |                                                                                                                                                                               |
|                        | arch02.uc24                                               | 4.6.0            |                                                                                                                                                                               |
|                        | arch02.engineering.uc24.firstboot                         | 4.6.0            |                                                                                                                                                                               |
|                        | Hardware Support X3-4.6.0<br>(arch01.hw)                  | 4.6.0            |                                                                                                                                                                               |
|                        | Linux kernel X3-4.6.0+6.12.40-ctrlx-55<br>(arch01.kernel) | 6.12.40-ctrlx-55 |                                                                                                                                                                               |
|                        | Hardware Support X7-4.6.0<br>(arch02-hw)                  | 4.6.0            |                                                                                                                                                                               |
|                        | Linux kernel X7-4.6.0+6.12.40-ctrlx-55<br>(arch02.kernel) | 6.12.40-ctrlx-55 |                                                                                                                                                                               |
|                        | Hardware Support OS-4.6.0<br>(arch03-hw)                  | 4.6.0            |                                                                                                                                                                               |
|                        | Linux kernel OS-4.6.0+6.12.40-ctrlx-54<br>(arch03.kernel) | 6.12.40-ctrlx-54 |                                                                                                                                                                               |
|                        | Automation Core-4.6.0<br>(automationcore)                 | 4.6.0            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Resolved defects and modifications</li> <li>● ↗ Note on use and known restrictions</li> </ul>           |
|                        | Device Admin-4.6.0<br>(deviceadmin)                       | 4.6.0            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Resolved defects and modifications</li> <li>● ↗ Note on use and known restrictions</li> </ul>           |
|                        | Version Guard-4.6.0<br>(gating)                           | 4.6.0            |                                                                                                                                                                               |
|                        | Setup-4.6.0<br>(setup)                                    | 4.6.0            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Note on use and known restrictions</li> </ul>                                                           |
|                        | Solutions-4.6.1<br>(solutions)                            | 4.6.1            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Resolved defects and modifications</li> </ul>                                                           |
|                        | bare-1.0<br>(bare)                                        | 1.0              |                                                                                                                                                                               |
|                        | core24-20260107<br>(core24)                               | 20260107         |                                                                                                                                                                               |
|                        | snapd-2.72<br>(snapd)                                     | 2.72             | ↗ <a href="https://github.com/snapcore/snapd/blob/master/packaging/ubuntu-16.04/changelog">https://github.com/snapcore/snapd/blob/master/packaging/ubuntu-16.04/changelog</a> |
| XOH-4.6.0              | XOH                                                       | 4.6.0            |                                                                                                                                                                               |
| XOV-4.6.0+<br>balanced | XOV                                                       | 4.6.0+balanced   |                                                                                                                                                                               |

### 3.1.2 Compatibility with drive-integrated control

In case of a drive-integrated control, ensure that the control version matches the drive firmware version.

The following table shows the combination options:

| Control XCR        | Drive firmware               |
|--------------------|------------------------------|
| 1.12               | AXS-V-0306                   |
| 1.12.15 and higher | AXS-V-0308                   |
| 1.14               | AXS-V-0308                   |
| 1.16               | AXS-V-0312                   |
| 1.18               | AXS-V-0314                   |
| 1.20               | AXS-V-0316 and AXS-V-0402    |
| 2.2                | AXS-V-0404                   |
| 2.4                | AXS-V-0502                   |
| 2.6                | AXS-V-0504                   |
| ≥ 2.6.10           | ≥ AXS-0504.4<br>≥ AXS-0606.0 |
| 3.2                | AXS-V-0602                   |
| 3.4                | AXS-V-0604                   |
| 3.6                | ≥ AXS-0504.4<br>≥ AXS-0606.0 |
| 4.2                | AXS-V-0608                   |
| 4.4                | AXS-V-0610                   |
| 4.6                | AXS-V-0612                   |

## 3.2 New functions

### 3.2.1 Automation Core

#### Feature 1114792: Pending information messages in alarms

##### Description

This functionality provides the option of displaying pending information messages in alarms.

This allows the user to be informed of non-critical information that is neither a warning nor an error.

- Pending information messages are displayed under “*Diagnostics* → *Logbook* → *Alarms*” and in the header of ctrlX OS (bell) with a blue information symbol.
- Pending information messages have to have the diagnostic class "A" and the priority "6" -> e.g. 080A6FFF.
- Pending information messages are always in the "reset" status. The "Reset" action is not supported.
- Pending information messages can always be deleted and then disappear.

New interfaces in the diagnostic system:

- Data Layer node:
  - *diagnosis/events/message-set-A6* -> The event counter of this node is incremented when the specific event "Diagnostic message with priority 6 set" occurs.
  - *diagnosis/get/actual/list/amount-Infos* -> This node provides the functionality to retrieve the number of outstanding information/messages in the current list with all active and reset diagnostics.
- C-interface:
  - *getAmountOfInfos()* -> This method returns the number of pending information/messages in the alarm list with all active and reset diagnostics.
- Diagnostics Distributor:

- Diagnostic event "MESSAGE\_SET\_A6" -> The event is triggered when a diagnostic information/message with priority 6 occurred. Can be used with the callback parameter types `CallbackParametersType::BASE`, `CallbackParametersType::COMMON` and `CallbackParametersType::COMMON_ENTITY`.

### Feature 1114982: Changes in the Data Layer

#### Description

New functionality:

- Realtime copy: Copies data between real-time memories when an owner output calls `beginAccess()`.
- Trace for `comm.datalayer.client-request`: Shows which request was sent and which response was received.
- `DLR_clientSubscribeModifySync()` and `DLR_clientSubscribeModifyASync()` added to the C-interface.
- Target platform: `armhf`

Bugfixes:

- Incompatible change: DL RT now works between 32- and 64-bit applications. This is an incompatible change for external applications. If you have an external application that uses DL RT, you will need to rebuild your application to work with `app.automationcore` 4.6 and higher. You do not need to create the application again if it runs as bundle within `app.automationcore`.
- Aliases whose source contains escape characters were not deleted correctly after removal.
- Only a single, unambiguous error message is now sent for a subscription.
- SONAME of the shared library is now only the main version (e.g. `lib-comm.datalayer.so.3`).
- Possible race condition when calling `ISubscription::getNodes()` fixed.
- Possible race condition when calling `setError` for a program task fixed.
- `getAuthToken` of the client no longer incorrectly returns an empty string during IPC communication.

### Feature 1131434: Linux Kernel Update

#### Description

- Bugfix: `trace_printk` warnings removed from log
- Support for AMD XDNA drivers added
- Hilscher netX driver updated to current version

## 3.2.2 Device Admin

### Feature 282405: DHCP server support in ctrlX OS

#### Description

Support for DHCP server added in ctrlX OS.

If ctrlX is used as gateway for connected devices, IP addresses can now be automatically assigned to devices via DHCP.

The DHCP server can be configured via the ctrlX web interface, enabling simple network setup for devices connected to ctrlX.

### Feature 1041322: "Apps" and "Licenses" were merged

#### Description

The "Apps" and "Licenses" tiles in "Settings" and their respective sites was merged into a single "Apps & Licenses" tile.

It is now also possible to install ZIP archives, which can contain both apps and a license file. Such archives can be downloaded from the ctrlX Device Portal for example.

#### **Feature 1051992: Support for embedding third-party applications in the ctrlX user interface via iFrame**

##### **Description**

- Consistent user experience:  
Enables seamless embedding of third-party applications into the ctrlX user interface using iFrames.
- Manifest schema extension:  
The package manifest schema has been updated to include a new embedded value for the target property.
- Routing integration:  
A new routing pattern /frame/<target-url> and EmbeddedFrameComponent has been introduced to handle the binding of the iFrame source.
- Weblib.core improvements:  
PackageManifestService has been extended to resolve embedded targets and manage the logic of the active sidebar menu for integrated applications.

#### **Feature 1058176: Creating a user account during initial commissioning**

##### **Description**

Create a user account during initial commissioning. Enter the desired username and password as well as the password confirmation, accept the "General Terms of Use" and click on "Create". You are then logged in.

Make a note of the username and password. These are absolutely required to log in again.

The new user has administrator rights and replaces the default user configured upon delivery.

#### **Feature 1155681: Public keys can be copied to the clipboard**

##### **Description**

In certificate management, you can display the public part of a key in any format (PEM or OpenSSH) and copy it to the clipboard with one click.

### **3.2.3 Setup**

#### **Feature 1112318: Warning of downgrade**

##### **Description**

To avoid accidental downgrades, the "Restore" page warns users if a backup file contains apps with an older major version. Users can then use the "Customized" option to check the affected apps and exclude them from the recovery process.

#### **Feature 1112320: Support for additional settings**

##### **Description**

The following settings are also supported:

- DHCP server settings
- Settings to synchronize licenses (License Center)

### 3.2.4 Solutions

#### Feature 995994: Extended Git functionality

##### Description

- Direct versioning of the active configuration
- Simplified SSH key management for Git remotes
- Simplified comparison of binary files
- Status display for uncommitted changes
- Support for creating and deleting Git tags
- Configurable fetch behavior for remotes
- Option to change commits (Amend)

## 3.3 Resolved defects and modifications

### 3.3.1 Automation Core

#### Bug 1091116: Bugfix for sporadic startup problem due to unstable eMMC enumeration

##### Description

A boot error occurred sporadically on the ctrlX CORE X3, during which the boot process was aborted. The cause was an inconsistent assignment of the device nodes for the internal eMMC memory by the Linux kernel. Depending on the detection sequence at startup, the internal memory was mounted either as /dev/mmcblk0 or /dev/mmcblk1. The system was configured to expect the internal memory explicitly under /dev/mmcblk0. If the memory was mounted as /dev/mmcblk1 instead, the access failed, which led to an automatic reboot of the device after a timeout of 60 seconds.

##### Troubleshooting

The problem was solved by implementing a permanent alias in the device tree of the system. This entry assigns a fixed ID to the eMMC controller, which is responsible for the internal memory. This ensures that the Linux kernel registers the internal memory reliably and consistently as /dev/mmcblk0 upon each system start. The unstable enumeration is thus prevented and the device can now boot reliably.

#### Bug 1098779: System message "Cannot find unit for notify message of PID \*\*\*\*, ignoring." in the logbook

##### Description

When activating the system messages in the logbook, there are many extra entries with the text "Cannot find unit for notify message of PID \*\*\*\*, ignoring."

#### Bug 1100451: System image update from version 2.6 and older to version 4.4 via the ctrlX OS web interface results in an error

##### Description

A system image update from version 2.6 and older to version 4.4 via the "Firmware Update" window in the ctrlX OS web interface causes an error and is therefore not possible.

##### Troubleshooting

The Rescue System component in the system image was updated to version 1.1.14+hotfix.3.

### **Bug 1116120: Backup/restore may fail after 30 days operating time of the device**

#### **Description**

Backup/restore and requires an app-specific temporary folder. This folder is created upon startup. With CORE24, a change was introduced with regard to cleaning up temporary folders. With the current version used by ctrlX OS, temporary folders that are not used for 30 days are removed. After removal, an app can no longer access its app-specific temporary folder, which can cause various errors, e.g. backup/restore can no longer fulfill its task, apps can no longer be installed, etc.

### **Bug 1126017: Frequent error 081F0042/0C7B4285 when switching from OPERATING to SERVICE**

#### **Description**

When switching from OPERATING to SERVICE, the error "081F0042 Error in Scheduler system / 0C7B4285 Switching state returned 'SCHED\_E\_LOCK\_TIMEOUT'" can occur often. Confirm this error with PROCEED. Correct switching despite error message.

### **Bug 1139268: Extended time synchronization (e.g. PTP) resulted in jitter in real-time processes**

#### **Description**

If the new function for time synchronization (without its own user interface, does not affect NTP in the "Date & Time" window) was active, there could be interruptions or delays (jitter) in real-time processes.

## **3.3.2 Device Admin**

### **Bug 1076361: Time and time zone are not applied**

#### **Description**

Ubuntu Core 24 no longer supports some time zones (e.g. Asia/Calcutta). However, these outdated time zones can still be selected in the web interface. If such a time zone is selected, the time information is not applied.

### **Bug 1116146: ctrlX OS sporadically takes longer to boot on ctrlX CORE**

#### **Description**

Updating the kernel from 6.1 to 6.12 can cause the internal eMMC controllers on a ctrlX CORE to be initialized in the "wrong" order. The internal memory is connected to one of the two controllers, the other controller is not connected. If the unconnected controller is initialized first, the name of the internal storage medium changes from /dev/mmcblk0 to /dev/mmcblk1. Ctrlx-vault is then persistently mounted on this device. Upon the next restart, snapd tries to mount the ctrlx-vault on this device. However, if the connected controller is faster during this startup, it gets the name /dev/mmcblk0 and the mounting will freeze for 90 seconds before the control finishes starting.

### **Bug 1124644: Powercontroller firmware is not updated**

#### **Description**

If an update of the FPGA and the power controller firmware are carried out simultaneously on an X5/X7 (e.g. by upgrading from v3.6 to v4.6), an error message is displayed stating that the power controller has the wrong firmware version after the upgrade.

### Troubleshooting

The firmware of the power controller is only updated if it is briefly disconnected from the power supply. There is a special system reboot for this, which was also executed if only the power controller was updated. However, if both the FPGA and the power controller are updated, a simple reboot was carried out. This was fixed by selecting the reboot with voltage cycle.

#### Bug 1126501: After updating all system apps on devices with image 2.6, "Upgrade ctrlX OS" (remodeling) cannot be executed

##### Description

On devices with the image 2.6, it was not possible to run "Upgrade ctrlX OS" (remodeling) after updating all system apps. It was aborted with an error message.

#### Bug 1129085: Image update does not work reliably on ctrlX CORE X5

##### Description

The internal connected eMMC controller is not reliably initialized as mmcblk0 on the ctrlX CORE X5 with the kernel 6.12 (included in image 4.4), but with a probability of 50% as mmcblk1.

For the image update, the file rescue.toml is written, which defines the target medium for the image.

If the mmcblk1 medium is now detected as the boot medium before the image is updated, this is also written to rescue.toml and the image update cannot be carried out.

#### Bug 1133074: Memory leak in UPNP service

##### Description

There was a memory leak in the UPNP service that continuously increased memory consumption.

#### Bug 1134469: Logbook missing from system report

##### Description

Depending on the user authorizations, the logbook folder from the system report was missing.

### 3.3.3 Solutions

#### Bug 1122488: Incomplete backup of Git settings

##### Description

The settings for the SSH remote server have not been backed up and restored.

## 3.4 Notes on use and known restrictions

### 3.4.1 Updating the apps on the ctrlX CORE Virtual

The setup ctrlX WORKS provides a new device ctrlX CORE Virtual compatible with the current version, including the current system apps. To convert an existing application to this version, proceed as follows:

**Execute the following steps in the described sequence:**

1. ➤ First connect to the ctrlX CORE Virtual and go to menu item "Manage app data" via  in the header or via *"Home" → tile "Manage app data"*.
2. ➤ Back up the current app data via  "Save to archive".

3. Save all available configurations by highlighting the respective configuration in the list and click on the button .
4. Create a new ctrlX CORE Virtual in ctrlX WORKS in the "Devices" window.
5. Install the application apps in ctrlX CORE.
  -  The application apps can be installed in the "Operating" state as long as not real-time apps (Motion, EtherCAT, PLC, PROFINET Device) are installed.  
When a real-time app is installed or available on the system, switch the "Operating" state to "Service" before the installation.
  - First install all application apps via the button  "Install from file".
  - Highlight all application apps and confirm them in the following file drop-down box. Subsequently, these apps are bundled and loaded to the control via "Install".
  - Wait until loading is completed.
  - Now press the button "Update all apps". The apps are installed.
  - Switch to the state "Service".
  - Update all real-time apps (Motion, EtherCAT, PLC, PROFINET Device).
  - Click on "Operating".
  - Subsequently, refresh the page to display the new system state.  
The control is ready again for operation.
6. Go to "Manage app data" and upload the relevant configuration backups via the  button. Subsequently, load the configuration to be executed via the  button. The update is completed.
7. Please verify that all data has been applied.

Further information can be found in the chapter [➔ Configurations – Introduction](#), ctrlX OS, Operating system for ctrlX CORE Control Devices, Application Manual.

### 3.4.2 Updating the apps on ctrlX CORE

Check the version of the installed apps on the ctrlX CORE control.



The following instructions are only valid if at least release version ctrlX CORE at least the release version XCR-V-0202 or a subsequent software version is installed.

**Preparation:** The apps to be updated were copied to a local directory on the PC on which the update can be executed.

The system apps can be installed in the "Operating" state as long as no real-time apps (Motion, EtherCAT, PLC, PROFINET Device) are installed.

As soon as a real-time app is installed or is available in the system, the status has to be changed from "Operating" to "Service" before installation.

**Execute the described steps in the described sequence:**

1. First connect to the ctrlX CORE and go to menu item "Manage app data" via  in the header or via "Home" → tile "Manage app data".
2. Back up current app data via  "Save to archive".
3. Save all available configurations by highlighting the respective configuration in the list and click on the button .
4. Select the menu item "Settings → Apps" to go to the overview of the installed apps.

**5.** Install the system apps now.

- Install all system apps via the button  “Install from file”.  
However, if the apps are installed individually, the following sequence must be followed:
  1. Update the "snapd" system app.
  2. Install the "core24" system app.
  3. Update the "Device Admin" system app.
  4. You can then install or update all other apps using the “Install from file” button.
- In the following file selection window, select all the desired apps and confirm your selection. Subsequently, these apps are bundled and transmitted to the control.
- Wait until loading is completed.
- The update information for the apps already available on the control and the new apps to be installed are displayed in a notification window.  
Press the button “Install” and confirm the dialog.
- Acknowledge the following messages with “Continue”. This starts the installation.
- After the installation, log into the ctrlX CORE again or go to the “Apps” window.
- Subsequently, refresh the page to display the new system state.
- The update of the system app has been completed.

**6.** Install the required application apps.

The next steps are always required, even if the system apps have been installed via the rescue system or you have received a control on which the current system apps are already installed. If the control is not yet in this state, first press the “Service”).

- Install all application apps and real-time apps (Motion, EtherCAT, PLC, PROFINET Device) via the button  “Install from file”.
  - In the following file selection window, select all the desired apps and confirm your selection.
  - Press the button “Install” and confirm the dialog.
  - Acknowledge the following messages with “Continue”. This starts the installation.
- ➔ The control is ready for operation again.

**7.** Go to “Manage app data” and upload the relevant configuration backups via the  button. Subsequently, load the configuration to be executed via the  button.

- ➔ The update is completed.

### 3.4.3 Automation Core

**Bug 589936: X3: Retain memory > 56 kB is sporadically inconsistent after switching off/on with drive-integrated control and ctrlX CORE X3**

**Description**

If more than 56 kB of retain data is used for the ctrlX CORE application(s) (e.g. in PLC applications), this may lead to an incorrect saving of retain variables stored in this area above the 56 kB limit when switching off.

**Effect:**

As a result, incorrect retain data is restored the next time the control is switched on.

**Affected products:**

The behavior can occur with the hardware variants ctrlX CORE X3 in ctrlX DRIVEplus (FD before 18/22) as well as ctrlX CORE X3 (FD before 01/22).

**Workaround**

Do not use more than 56 kB of retain data with the hardware variants mentioned.

**Bug 711581: Device does not start with USB flash drive inserted on XF01C****Description**

The USB port XF01C has a higher boot priority than the internal hard disk. This causes the boot process to deviate from the internal hard disk and the operating system on the USB flash drive tries to boot.

If the boot process deviates from the expected path, for example, by booting from an unsigned USB flash drive instead of the internal hard drive, Secure Boot can block the boot process as it could pose a potential threat to system integrity.

**Workaround**

Disconnect the USB stick from port XF01C and restart the device

**Bug 853922: ctrlX CORE X5: Device does not start sporadically if an external data carrier is plugged into port XF01A, XF01B or CF01****Description**

External data carriers at port XF01A, XF01B (USB sticks) or CF01 (microSD card) are sporadically recognized as boot media by the BIOS of the ctrlX CORE X5. The result is that the boot process deviates from the internal hard disk and the operating system on the external medium tries to restart.

If the boot process deviates from the expected path, for example by booting from an unsigned USB flash drive or microSD card instead of the internal hard disk, Secure Boot can block the boot process as it could pose a potential threat to system integrity.

**Workaround**

In the event of an error, briefly remove the external data carrier before the system starts up and plug it back in during runtime.

**Bug 867074: Packet loss during redundant communication with HSR****Description**

The HSR protocol for redundant network communication bundles two physical network interfaces into a single virtual network interface. The resulting virtual network interface is used to address the redundant network communication. In the current implementation, a package loss of < 1% was observed when using HSR to bundle the network interfaces of ctrlX CORE X3, i.e. XF10, XF50 and XF51. However, no packet loss occurred when using the network interfaces of ctrlX CORE X7, i.e. XF10, XF11 and XF12.

**Bug 1014687: ctrlX CORE X5/X7: First boot after image update (hardware reset) takes longer****Description**

During the first boot after the image update, it may take up to 10 minutes for the device to be accessible via the network. However, the image installation was carried out successfully. Only the first initial start-up of the device after the image update is affected. All other boot operations are not affected by this.

**Workaround**

Connect the Ethernet ports XF10, XF11 and XF12 to the network during the initial start-up. Each port has to be assigned an IP address automatically, for example via DHCP.

**Bug 1179239: Interrupting real-time tasks at full core load****Description**

Long-running real-time tasks that fully utilize a processor core may be interrupted once a second.

**3.4.4 Device Admin****Bug 899934: Uninstallation takes a very long time****Description**

If an app is uninstalled, this process takes a very long time. The error is in the underlying operating system.

**Bug 1021621: Hardware support cannot be updated on a control with Secure Boot (ctrlX CORE X5/X7) from the store****Description**

The attempt to update the Hardware Support App from the store fails as the SBAT version, which is important in connection with Secure Boot, cannot be determined. The installation from the store is therefore rejected.

**Bug 1111609: Updating all apps on a device with image version 2.6.10 aborts sporadically after the core app****Description**

When trying to update all apps on a device with ctrlX OS image version 2.6.10 to a newer version via batch upload, the process stops sporadically after the Core app.

**Workaround**

First update the Core app. All other apps can then be updated together.

**Bug 1188123: Resetting the device data fails****Description**

After a runtime update to 4.6 RC2 and the installation of apps (e.g. core22, EtherCAT Master), "Reset Device Data" fails. Although the apps are included in the backup, they are not restored. The process terminates with a "502 Bad Gateway" error as an Automation Core cannot be reached to query the scheduler status.

### Workaround

Log in to the browser on the device. Then open the site "*Settings → Backup & Restore*" and click on the "Restore" tile. The modal dialog "Select backup file" opens. Click on (select) the backup file displayed in the dialog and click on the "Select" button. The modal dialog is closed and a new site opens in the background. In the "Device" section, select "Merge" and click on the "Start" button to start the restoration.

#### Bug 1189755: Uninstall button for the system app "core24" after remodeling

##### Description

After remodeling from core22 to core24, core24 becomes a "system app". After the update, an uninstall button is erroneously displayed in the app overview at core24.

#### Bug 1192160: Creating a system report seems to freeze

##### Description

Adding the logbook to a system report can take a long time (more than 20 minutes) if there are a lot of logbook entries. As no progress is displayed when adding the logbook, it may appear that the creation of the system report is freezing.

##### Workaround

Please wait until the logbook has been added to the system report (may take more than 20 minutes).

### 3.4.5 Setup

#### Bug 1192174: Restoring a backup on a control of a different type fails

##### Description

For example, if you have created a backup including system apps on a virtual control and want to restore it to a real target device with the same processor architecture (AMD, e.g. X7), the restore fails when trying to update hardware-specific apps (Hardware Support, Realtime Kernel).

##### Workaround

On the "Restore" page, enable the "Customized" option. Deselect the apps "Hardware Support" and "Virtual Kernel" or "Realtime Kernel" if these cannot be installed on the target device.

## 4 Version 4.4.1

### 4.1 Software



This release is part of ctrlX CORE 4.4.0.

## 4.1.1 Components

Table 2: Included components

| Short name        | Software                                                  | Version          | Additional notes                                                                                                                                                                |
|-------------------|-----------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ctrlX Apps</b> |                                                           |                  |                                                                                                                                                                                 |
| XCR-4.4.1         | arch01.firstboot                                          | 4.4.1            |                                                                                                                                                                                 |
|                   | arch02.uc24                                               | 4.4.1            |                                                                                                                                                                                 |
|                   | arch02.engineering.uc24.firstboot                         | 4.4.1            |                                                                                                                                                                                 |
|                   | Hardware Support X3-4.4.0<br>(arch01.hw)                  | 4.4.0            |                                                                                                                                                                                 |
|                   | Linux Kernel X3-4.4.0+6.12.40-ctrlx-52<br>(arch01.kernel) | 6.12.40-ctrlx-52 |                                                                                                                                                                                 |
|                   | Hardware Support X7-4.4.0<br>(arch02-hw)                  | 4.4.0            |                                                                                                                                                                                 |
|                   | Linux Kernel X7-4.4.0+6.12.40-ctrlx-52<br>(arch02.kernel) | 6.12.40-ctrlx-52 |                                                                                                                                                                                 |
|                   | Hardware support OS-4.4.0<br>(arch03-hw)                  | 4.4.0            |                                                                                                                                                                                 |
|                   | Linux Kernel OS-4.4.0+6.12.40-ctrlx-52<br>(arch03.kernel) | 6.12.40-ctrlx-52 |                                                                                                                                                                                 |
|                   | Automation Core-4.4.5<br>(automationcore)                 | 4.4.5            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Bugfixes and modifications</li> <li>● ↗ Note on use and known restrictions</li> </ul>                     |
|                   | Device Admin-4.4.0<br>(deviceadmin)                       | 4.4.0            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Bugfixes and modifications</li> <li>● ↗ Note on use and known restrictions</li> </ul>                     |
|                   | Version Guard-4.4.0<br>(gating)                           | 4.4.0            |                                                                                                                                                                                 |
|                   | Setup-4.4.0<br>(setup)                                    | 4.4.0            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Bugfixes and modifications</li> <li>● ↗ Note on use and known restrictions</li> </ul>                     |
|                   | Solutions-4.4.0<br>(solutions)                            | 4.4.0            | <ul style="list-style-type: none"> <li>● ↗ New functions</li> <li>● ↗ Bugfixes and modifications</li> <li>● ↗ Note on use and known restrictions</li> </ul>                     |
|                   | bare-1.0<br>(bare)                                        | 1.0              |                                                                                                                                                                                 |
|                   | core22-20250612<br>(core22)                               | 20250612         |                                                                                                                                                                                 |
|                   | core24-20250729<br>(core24)                               | 20250729         |                                                                                                                                                                                 |
|                   | snappd-2.71<br>(snappd)                                   | 2.71             | ↗ <a href="https://github.com/snapcore/snappd/blob/master/packaging/ubuntu-16.04/changelog">https://github.com/snapcore/snappd/blob/master/packaging/ubuntu-16.04/changelog</a> |
| XOH-4.4.0         | XOH                                                       | 4.4.1            |                                                                                                                                                                                 |
| XOV-4.4.0         | XOV                                                       | 4.4.1+balanced   |                                                                                                                                                                                 |

### 4.1.2 Compatibility with drive-integrated controls

In the case of a drive-integrated control, ensure that the version of the control matches the version of the drive firmware.

The following table shows the possible combinations:

| XCR control        | Drive firmware               |
|--------------------|------------------------------|
| 1.12               | AXS-V-0306                   |
| 1.12.15 and higher | AXS-V-0308                   |
| 1.14               | AXS-V-0308                   |
| 1.16               | AXS-V-0312                   |
| 1.18               | AXS-V-0314                   |
| 1.20               | AXS-V-0316 and AXS-V-0402    |
| 2.2                | AXS-V-0404                   |
| 2.4                | AXS-V-0502                   |
| 2.6                | AXS-V-0504                   |
| ≥ 2.6.10           | ≥ AXS-0504.4<br>≥ AXS-0606.0 |
| 3.2                | AXS-V-0602                   |
| 3.4                | AXS-V-0604                   |
| 3.6                | ≥ AXS-0504.4<br>≥ AXS-0606.0 |
| 4.2                | AXS-V-0608                   |
| 4.4                | AXS-V-0610                   |

## 4.2 New functions

### 4.2.1 Automation Core

#### Feature 740428: Improved handling of Motion errors

##### Description

Clicking on the "Reset errors" button in the axis commissioning dialog resets the associated Motion object in the Motion application and all associated errors and warnings in the alarms dialog. The same applies when deleting an associated Motion error in the alarms dialog. Previously, resetting the Motion object and deleting the error in the alarms dialog had to be carried out as two separate actions.

#### Feature 870976: Functional extensions in the logbook web interface

##### Description

The user rights for the logbook have been revised and adapted to the general user rights of ctrlX. While the old user rights regulated the availability of log message types (diagnostics, trace and system), the new user rights focus on the availability of functionalities ("view", "operate" and "manage"). After updating the software, the old rights continue to exist, but are marked as "deprecated". As soon as a new user right is assigned, the old rights lose their validity.

The handling of Motion errors has been improved.

The display of the columns in the default display of the logbook can now also be configured.

After an error reset, a message is displayed confirming that the reset has been completed.

There is a bookmark filter in the history overview that can be used to display only those entries for which a bookmark has been set.

The bookmark icon has been edited to improve the visualization of the difference between "set" and "not set".

All filter dialogs have been revised and standardized.

The date in the date column is displayed in the format that corresponds to the language set.

When the date filter is opened, the date of the currently selected entry is automatically entered in the filter.

### **Feature 938847: Updating the Celix Framework**

#### **Description**

The Celix Framework has been updated to version 2.4.

### **Feature 984227: Updating the "core" system app**

#### **Description**

The "core" system app has been updated to core24. This version is based on the Ubuntu 24.04 release.

### **Feature 995017: Various functional adjustments to the Data Layer web interface**

#### **Description**

- Displaying the histogram data in a new area of the detailed view  
For all nodes that have the "readType" "histogram", a new additional area is displayed in the detail view in which the data from the node is displayed in a chart diagram.
- Creating "raw" data is now possible  
For nodes that have the "createType" "raw", creation with any files is possible.
- Creating "raw" data is now possible  
For nodes that have the "createType" "raw", creation with any files is possible.
- Escape characters in the node names and addresses are taken into account  
Addresses and names of nodes can now contain an escape character ('\'). These are filtered out by the interface for display.
- Nodes with "eventType" can now also be subscribed to the Data Layer for reading the values
- Time stamps of the data can also be displayed in the input field  
The display of the time stamp for the data can be enabled via a setting in the settings dialog.
- Optional scalars in JSON schema are now supported

### **Feature 996130: Changes in the Data Layer**

#### **Description**

New functionality:

- Added environment variable DL\_DEBUG to enable the debug output
- Task::onMetadata() added to change the meta data of a program task
- Version for 32-bit Linux
- InheritMutex and InheritRecursiveMutex are obsolete, use common::aos variants instead
- Events and diagnostics for connecting/disconnecting in devices/remotes
- Security enabled when used as a bundle

- Security can be enabled by defining certificate paths
- Escaping for addresses with ``
- ProgramTask now has the fields createdAt and finishedAt
- New flatbuffer version used (oss.flatbuffers 3.0.0+25.2.10)
- The static Data Layer library now contains all the required static libraries
- Remote Procedure Call (RPC)
- Removed dependency of trace and systemd libraries from comm.datalayer libraries
- Automatic provision of sub-nodes
- Websocket support
- Reverse connections are now enabled by default

#### Bugfixes:

- tokencheck.enable in celix config.properties did not work
- address2.h: escape() and unescape() are now publicly static
- RPC crash if no shared memory can be created
- SubnodeProvider:
  - onWrite did not work with asynchronous providers
  - Incorrect metadata for leaf nodes (cannot be browsed)
- Removing nodes of the subscribed list before calling onUnsubscribe the subscription nodes
- Static library now contains all required libraries (except openssl for security reasons)
- Thread name for identity change
- Possible crash if areas are registered via mddb
- Allowing the creation of areas for Asterisk nodes
- Possible crash if 8-bit ANSI is used as username/password in the connection string
- Improving the memory utilization of the Compliance Checker
- Possible crash when calling Variant::deserializeShared()
- Collision of shared memory areas in Windows
- openMemory() did not wait for the first connection attempt to the RT memory by default
- Possible crash if a variant with invalid string termination is received
- No subscription data on the remote node (Bug 835720)

### Feature 1045881: Data Layer improvements

#### Description

##### New functionality:

- Loss-free subscription of entire rt/data nodes
- Nodes of RT link to master RT nodes in rtNode reference
- Writing shared retain data with Data Layer nodes
- findMDDb in C-interface
- The affinity of threads (providers and clients) can be configured in *framework/affinity/nrt-communication*
- Add *datalayer/statistics/provider/<id> /mddb* to display the provider's MDDBs
- Supporting the registration of subnodeprovider with bfbs instead of type strings

##### Bugfixes:

- No automatic reconnection of TCP client and providers when the broker is restarted
- Multiline description in the subnode provider
- Always inform if there is no connection established when waiting in provider::start()
- Failed start/stop of the provider when using thousands of nodes
- NVRAM synchronization counter only increases when NVRAM is actually updated

#### Feature 1047945: Flexible core division

##### Description

Below the Data Layer node *framework/affinity*, the core can be divided for the following areas:

- NRT communication
- Scheduler
- PLC Runtime

Compared to previous versions, threads can now use all cores for communication in the standard configuration and no longer just the first RT core. This means that communication with the control system is still possible in many applications despite the high system load.

#### Feature 1048023: Linux Kernel updated to version 6.12.40

##### Description

Update of the following components:

- Linux Kernel 6.12.40
- HailoRT v4.23.0
- NVIDIA Linux Open GPU Kernel Module 580.65.06

#### Feature 1059916: Functional extensions in the Data Layer web interface

##### Description

Several extensions have been implemented in the Data Layer web interface:

The "Data Layer node" widget is available on the start page, which can be used to display a value from a Data Layer node. Possible nodes that can be displayed in the widget have to be of type "number", "string" or "boolean". The widget display can be customized by the user depending on the node type.

Various dialogs in the Data Layer web interface (filter dialog for selecting big data nodes, dialog for creating child nodes) can be individually resized by the user.

All integer values (signed and unsigned) are displayed according to the set display format. The following formats (in addition to the decimal format) are possible: Bin, Oct and Hex. These integer values can also be edited in the writable input fields in all possible formats (decimal, bin, oct and hex). After editing, however, they are displayed again according to the set display format.

Online help can be called up for a node if the corresponding link is available in the meta data.

The node description is now displayed in multiple lines if the description contains a "\n" in the relevant places.

## 4.2.2 Device Admin

### Feature 548599: Balanced energy mode for ctrlX OS (Balanced Mode)

#### Description

Two images are now provided for virtual devices. An image with which the virtual control is always operated at maximum energy utilization (performance mode). Furthermore, a second image is provided for virtual devices, which is operated with balanced energy utilization (balanced mode).

The switching function between the two modes is not yet implemented, but will be available in a future version of ctrlX OS.

### Feature 836887: Device Admin supports CAN interfaces

#### Description

If a CAN interface is connected to a device, its baud rate can be set in the network settings and via a Data Layer command.

### Feature 914768: Mounting and unmounting storage media using Data Layer commands

#### Description

Removable storage media can be mounted and unmounted via Data Layer commands.

### Feature 948328: Simple Network Management Protocol can be used in ctrlX OS

#### Description

Simple Network Management Protocol (SNMP) is a network protocol that is used to monitor devices in a network. It enables the collection of information about the network device status.

### Feature 986338: Notification for differing time

#### Description

If the ctrlX OS time differs from the browser time, this is displayed in the interface via a notification banner. The display continues across all pages until the time is correct again.

### Feature 986818: Supporting VLAN via DAN network interfaces

#### Description

VLAN network interfaces with DAN network interfaces can be created as parent nodes.

### Feature 996303: System report

#### Description

The system report contains information on the file system (Storage Media) and the RAM in the device-info.json file.

### Feature 1006016: Renaming "Firmware update" to "Flash system image"

#### Description

The "Firmware update" tile under "Settings" has been renamed to "Flash system image".

### Feature 1032629: Supporting Single Sign-on (SSO)

#### Description

It is now possible to log in to ctrlX OS using single sign-on via an identity provider. ctrlX OS has to be configured accordingly.

**Feature 1040678: Improved display of the login banner for ctrlX OS****Description**

When starting ctrlX OS, users are greeted with a terminal screen and a banner on the connected display.

All relevant network interface information is now displayed, which improves accessibility and user-friendliness.

- Displays all active IP addresses (IPv4 and IPv6) per interface.
- Simplifies the device connection from any network.
- Improves user-friendliness in virtual and multi-interface setups.

**Feature 1054708: 10-day trial period available for selected apps****Description**

Some apps, such as Firewall, Security Scanner and VPN Client, now support the one-time activation of a 10-day trial period directly under ctrlX OS.

**Feature 1060885: Displaying the ctrlX OS version in "System information"****Description**

The ctrlX OS version is set under *""Settings" → "System information" "* in the "General" section.

**Feature 1061268: Display of remaining time in trial mode****Description**

A ctrlX CORE virtual or an unlicensed installation of ctrlX OS switches off automatically after 4 hours. The time remaining until switch-off is displayed in the header.

**Feature 1071645: Display CPU information in "System information"****Description**

The CPU information is displayed under *""Settings" → "System information" "* in the "Hardware" section.

### 4.2.3 Setup

**Feature 981999: Deleting backup files after restoring****Description**

Backup files are deleted from the device after data has been restored successfully.

**Feature 1048209: Supporting additional settings****Description**

The following settings are also supported:

- CAN network settings
- Settings for the SNMP server
- Settings for Single Sign-On (SSO)
- Configuring network drives (NFS, SMB)

### 4.2.4 Solutions

**Feature 995857: Configurations can be added to version control with Git****Description**

Configurations can be version managed with Git.

The following functions are available:

- Adding archived configuration to Git
- Displaying and discarding local changes
- Tracking new and changed files
- Creating, checking out and comparing commits
- Show Git commit history
- Loading and saving the configuration taking into account the Git history
- Downloading, uploading and copying the configuration taking the Git history into account
- Managing Git settings on the device

#### **Feature 995988: Collaborative engineering with Git**

##### **Description**

Users can publish their changes in Git remote repositories. Branches can be used to manage commits and work together on technical tasks.

The following functions are available:

- Set up SSH connection
- Clone remote repositories
- Retrieving and transferring changes
- Creating and deleting branches
- Check out branches
- Merge branches
- Keeping an overview with the branch diagram

## **4.3 Resolved defects and modifications**

### **4.3.1 Automation Core**

#### **Bug 1010221: Error when creating the callable during start-up**

##### **Description**

In the PLC, you can use external events that automatically generate a callable in the ctrlX CORE Scheduler. During start-up, an error sporadically occurred when creating these callables.

#### **Bug 1117971: Error when automatically creating callables**

##### **Description**

When automatically creating callables, the error "Error during automatic reallocation of the callables" could occur.

### **4.3.2 Device Admin**

#### **Bug 940111: SD card can sporadically not be read in Windows PC after firmware update**

##### **Description**

If an SD card is partitioned via the partition management in the "Memory" interface, it is detected under Windows.

However, if the card is partitioned by means of a firmware update, the storage medium sporadically cannot be read under Windows.

##### **Bugfixing**

Before creating the new partition table, all existing partitions are first deleted. The SD card is subsequently also identified in the Windows PC.

**Bug 994012: "Upgrade ctrlX OS" (remodeling) fails sporadically****Description**

If "Upgrade ctrlX OS" is selected on the apps page, the error message "Remodeling execution failed" is reported sporadically.

**Bugfixing**

"Upgrade ctrlX OS" (remodeling) is repeated if it fails.

**Bug 995534: A USB flash drive cannot be mounted again if it is removed without clicking on "remove safely"****Description**

If you remove a mounted storage medium without clicking on "remove safely", the storage medium cannot be mounted again without restarting the device.

**Bugfixing**

If you remove a mounted storage medium without clicking on "remove safely", the storage medium can be re-inserted and mounted again.

**Bug 1043154: When formatting a partition, the name of the storage medium must be specified****Description**

According to the API, formatting a device using the "media" or "device" parameter is possible. Due to another bugfix, formatting was currently only possible with the "device" parameter.

**Bugfixing**

Formatting is now possible again, as specified, using the "media" or "device" parameters.

**Bug 1049110: Snapd app update fails for devices without serial assertion****Description**

The serial assertion is usually created automatically when the device is connected to the internet and authenticates itself with the ctrlX Store. If a device has not received a serial assertion since the last image update, an error occurs when updating the Snapd app.

**Troubleshooting**

Since then, the installation process has only waited a certain amount of time until the device is ready for installation. It is now ensured that the system waits until this state is reached.

**Bug 1064879: Alarm "Failed to set CPU configuration" is displayed****Description**

The alarm "Failed to set CPU configuration, preconditions not met" is incorrectly displayed as an error in the logbook. The reason for this is the change-over from cgroups-v1 to cgroups-v2.

#### **Bug 1079925: "Upgrading ctrlX OS" (remodeling) not possible with snapd 2.68.5.1**

##### **Description**

If ctrlX version 1.20 has been installed on a device and all system apps have been updated to version 3.6 (including snapd 2.68.5.1), "Upgrade ctrlX OS" is displayed on the "Apps" page. If you click on this, the error message "Remodeling execution failed" is displayed.

##### **Bugfixing**

Snapd 2.71 supports "Upgrade ctrlX OS" (remodeling) again.

#### **Bug 1081521: The system operating time is not updated in the Data Layer**

##### **Description**

The system operating time is only updated once when the control unit is started in the Data Layer.

##### **Bugfixing**

The system operating time is continuously updated in the Data Layer.

#### **Bug 1126921: Error when restoring a subscription (certificate management)**

##### **Description**

If an attempt is made to change an existing subscription as part of a restore operation for an existing subscription, this leads to an error 404 ("Subscription id not found").

##### **Bugfixing**

Subscriptions can be restored.

### **4.3.3 Setup**

#### **Bug 1015711: Resetting the default password is not possible**

##### **Description**

If a default password was set in the setup app, it could not be reset.

#### **Bug 1035162: Unauthorized access to sensitive data via security mechanisms**

##### **Description**

A vulnerability in the web application of the ctrlX OS setup mechanism facilitated an authenticated (low privileged) attacker to gain remote access to backup archives created by a user with elevated permissions. Depending on the content of the backup archive, the attacker may have been able to access sensitive data.

##### **Troubleshooting**

Downloading the backup file is only permitted for administrators with "Full access" authorization.

#### **Bug 1035172: Unauthenticated access to API endpoint**

##### **Description**

A vulnerability in the Task API endpoint of the ctrlX OS setup mechanism allowed a remote, unauthenticated attacker to access and extract internal application data, including potential debug logs and the version of installed apps.

### Troubleshooting

The Task API provides only very limited information to unauthenticated users. Only users with "Full access" receive further information.

#### **Bug 1041006: Backup/restore does not support NFS and SMB mount points**

##### **Description**

When selecting an NFS or SMB mount point to create or restore a backup, an error occurred and the process terminated with an error message.

#### **Bug 1053299: Unencrypted backup files**

##### **Description**

Ambiguous wording in the web interface of the ctrlX OS setup mechanism could lead the user to believe that the backup file is encrypted when a password is set. However, only the private key code - if available in the backup - is encrypted, while the backup file itself remains unencrypted.

### Troubleshooting

"Encrypted ZIP file" is selected as the default format for backup files and the outdated description above the password field has been removed.

#### **Bug 1062953: Certain snapd versions cannot be restored**

##### **Description**

Certain snapd versions were not detected correctly when restoring a backup. As a result, snapd may be sporadically not updated and subsequent errors occurred, causing the recovery to fail.

#### **Bug 1068029: Error when restoring PKIs**

##### **Description**

PKIs that use a certificate that is not stored in the "Network Security" certificate store, but in a different one, such as "Data Layer" could not be restored.

#### **Bug 1071563: User-defined password policy is not deleted when restoring a backup**

##### **Description**

If a user ("user") was assigned a password policy ("named password policy") added by the user, and the user and the password policy were subsequently deleted by applying a backup, the error was reported that the password policy is in use and therefore cannot be deleted.

#### **Bug 1076429: User is not removed from group**

##### **Description**

If a user is a member of at least one group, but not part of a group in a backup file, and the backup is restored in "Restore" mode, the user is not removed from the group(s).

#### **Bug 1077634: Server-based licensing is not enabled when restoring a backup**

##### **Description**

Server-based licensing could be disabled by applying a backup, but not enabled.

#### Bug 1077648: Warning "Json schema issue: ..." when restoring a backup

##### Description

If an individual certificate and key are selected for client authentication ("individual certificate and key") in a backup with server-based licensing, two warnings, each starting with "Json schema issue: ...", are incorrectly issued when the backup is restored.

#### Bug 1079332: License server is not enabled when restoring a backup

##### Description

If the license server functionality ("floating license server") was disabled on a ctrlX device, the functionality could not be enabled by restoring a backup.

#### Bug 1113522: Certificate and PKI cannot be deleted together

##### Description

If a certificate is used in a PKI and both are not included in a backup, neither can be deleted by applying the backup in "Restore" mode.

### 4.3.4 Solutions

#### Bug 970580: Authorization "Manage configurations" is insufficient for loading and saving

##### Description

Depending on the app installation, authorization can be missing for users with the "Manage configurations" authorization, which results in failed loading or saving (403 Forbidden).

##### Troubleshooting

The "Manage configurations" authorization allows the user to load and save configurations (independently of other installed apps).

#### Bug 1070473: Uploading archives with read-only directories fails

##### Description

Configuration archives (ZIP files) could not be uploaded to a device if directories within the archive were read-only.

## 4.4 Notes on use and known restrictions

### 4.4.1 Updating the apps on the ctrlX CORE Virtual

The setup ctrlX WORKS provides a new device ctrlX CORE Virtual compatible with the current version, including the current system apps. To convert an existing application to this version, proceed as follows:

**Execute the following steps in the described sequence:**

1. First connect to the ctrlX CORE Virtual and go to menu item "Manage app data" via  in the header or via "Home" → *tile "Manage app data"*.
2. Back up the current app data via  "Save to archive".
3. Save all available configurations by highlighting the respective configuration in the list and click on the button .
4. Create a new ctrlX CORE Virtual in ctrlX WORKS in the "Devices" window.

#### 5. ➤ Install the application apps in ctrlX CORE.

 The application apps can be installed in the “Operating” state as long as not real-time apps (Motion, EtherCAT, PLC, PROFINET Device) are installed.

When a real-time app is installed or available on the system, switch the “Operating” state to “Service” before the installation.

- First install all application apps via the button  “Install from file”.
- Highlight all application apps and confirm them in the following file drop-down box. Subsequently, these apps are bundled and loaded to the control via “Install”.
- Wait until loading is completed.
- Now press the button “Update all apps”. The apps are installed.
- Switch to the state “Service”.
- Update all real-time apps (Motion, EtherCAT, PLC, PROFINET Device).
- Click on “Operating”.
- Subsequently, refresh the page to display the new system state.  
The control is ready again for operation.

#### 6. ➤ Go to “Manage app data” and upload the relevant configuration backups via the button. Subsequently, load the configuration to be executed via the button. The update is completed.

#### 7. ➤ Please verify that all data has been applied.

Further information can be found in the chapter ➔ [Configurations – Introduction](#), ctrlX OS, Operating system for ctrlX CORE Control Devices, Application Manual.

### 4.4.2 Updating the apps on ctrlX CORE

Check the version of the installed apps on the ctrlX CORE control.



The following instructions are only valid if at least release version ctrlX CORE at least the release version XCR-V-0202 or a subsequent software version is installed.

**Preparation:** The apps to be updated were copied to a local directory on the PC on which the update can be executed.

The system apps can be installed in the “Operating” state as long as no real-time apps (Motion, EtherCAT, PLC, PROFINET Device) are installed.

As soon as a real-time app is installed or is available in the system, the status has to be changed from “Operating” to “Service” before installation.

#### Execute the described steps in the described sequence:

1. ➤ First connect to the ctrlX CORE and go to menu item “Manage app data” via  in the header or via “Home” → *tile “Manage app data”*.
2. ➤ Back up current app data via  “Save to archive”.
3. ➤ Save all available configurations by highlighting the respective configuration in the list and click on the button .
4. ➤ Select the menu item “Settings → Apps” to go to the overview of the installed apps.

**5.** Install the system apps now.

- Install all system apps via the button  “Install from file”.  
However, if the apps are installed individually, the following sequence must be followed:
  1. Update the "snapd" system app.
  2. Install the "core24" system app.
  3. Update the "Device Admin" system app.
  4. You can then install or update all other apps using the “Install from file” button.
- In the following file selection window, select all the desired apps and confirm your selection. Subsequently, these apps are bundled and transmitted to the control.
- Wait until loading is completed.
- The update information for the apps already available on the control and the new apps to be installed are displayed in a notification window.  
Press the button “Install” and confirm the dialog.
- Acknowledge the following messages with “Continue”. This starts the installation.
- After the installation, log into the ctrlX CORE again or go to the “Apps” window.
- Subsequently, refresh the page to display the new system state.
- The update of the system app has been completed.

**6.** Install the required application apps.

The next steps are always required, even if the system apps have been installed via the rescue system or you have received a control on which the current system apps are already installed. If the control is not yet in this state, first press the “Service”).

- Install all application apps and real-time apps (Motion, EtherCAT, PLC, PROFINET Device) via the button  “Install from file”.
  - In the following file selection window, select all the desired apps and confirm your selection.
  - Press the button “Install” and confirm the dialog.
  - Acknowledge the following messages with “Continue”. This starts the installation.
- ➔ The control is ready for operation again.

**7.** Go to “Manage app data” and upload the relevant configuration backups via the  button. Subsequently, load the configuration to be executed via the  button.

- ➔ The update is completed.

### 4.4.3 Automation Core

**Bug 1051857: X3: Pending error after image update****Description**

During the image update, the error "Condition detected that triggered a safe machine state" can occur.

**Workaround**

A reboot fixes this error.

**Bug 1098779: System message "Cannot find unit for notify message of PID \*\*\*\*, ignoring." in the logbook****Description**

If you activate the system messages in the logbook, there are many entries in the logbook with the text "Cannot find unit for notify message of PID \*\*\*\*, ignoring.". This does not affect the functionality of the ctrlX and is only a message from the system.

**Bug 1100451: System image update from version 2.6 and older to version 4.4 via the ctrlX OS web interface results in an error****Description**

A system image update from version 2.6 and older to version 4.4 via the "Firmware Update" window in the ctrlX OS web interface results in an error and is therefore not possible.

**Workaround**

Download the ctrlX CORE system image from the collaboration room and create the USB flash drive for the firmware update using the "ctrlX OS Recovery" function in ctrlX WORKS.

**Bug 1111432: ctrlX CORE X5/X7: Warning 080E082A/0C64100D "App connection restored" after image update****Description**

With the ctrlX CORE X5/X7, the warning 080E082A/0C64100D "App connection restored - Missing app connection of the signed application is being restored" is displayed in the logbook after the image update upon the first startup. The reason for this is that an app interface has to be connected manually. After updating the image, the initial start-up is delayed by 10 to 20 seconds.

**Bug 1120682: ctrlX CORE X7 startup sporadically delayed by 140 seconds****Description**

The startup of a CORE X7 may be sporadically delayed by 140 seconds.

**Bug 1122688: Increased runtime of scheduler tasks****Description**

The maximum runtime of the scheduler tasks has been increased by up to 20% in version 4.4.0 compared to version 3.6.3.

**Bug 1126017: Frequent error 081F0042/0C7B4285 when switching from OPERATING to SERVICE****Description**

When switching from OPERATING to SERVICE, the error "081F0042 Error in Scheduler system / 0C7B4285 Switching state returned 'SCHED\_E\_LOCK\_TIMEOUT'" can occur frequently. This error can be confirmed with PROCEED. Switching takes place correctly despite the error message.

**Bug 1131271: Increasing "deviation" time measurement for scheduler tasks****Description**

The "Deviation" time measurement on a CORE X5/X7 results in an incremental increase in time.

#### 4.4.4 Device Admin

**Bug 855538: "Reset device data" can fail sporadically****Description**

When executing the setup, an error occurs which results in an error message after completing the operation.

**Bug 1021621: Hardware support cannot be updated on a control with Secure Boot (ctrlX CORE X5/X7) from the store****Description**

Attempting to update the Hardware Support App from the store fails as the SBAT version, which is important in connection with Secure Boot, cannot be determined. The installation from the store is therefore rejected.

**Bug 1076361: Time and time zone are not applied****Description**

Ubuntu Core 24 no longer supports some time zones (e.g. Asia/Calcutta). However, these outdated time zones can still be selected in the web interface. If such a time zone is selected, the time information is not applied. However, no error message is displayed.

**Workaround**

Use a supported time zone.

**Bug 1111609: Updating all apps on a device with image version 2.6.10 is canceled after the app Core22****Description**

When trying to update all apps on a device with ctrlX OS image version 2.6.10 via batch upload, e.g. to version 3.6.3, the process is canceled after the Core22 app.

**Workaround**

First update the core app. All other apps can then be updated simultaneously.

**Bug 1116146: ctrlX OS sporadically takes longer to boot on ctrlX CORE****Description**

Updating the kernel from 6.1 to 6.12 can cause the internal eMMC controllers on a ctrlX CORE to be initialized in the "wrong" order. The internal memory is connected to one of the two controllers, the other controller is not connected. If the unconnected controller is initialized first, the name of the internal storage medium changes from /dev/mmcblk0 to /dev/mmcblk1. Ctrlx-vault is then persistently mounted on this device. Upon the next restart, snapd tries to mount the ctrlx-vault on this device. However, if the connected controller is faster during this startup, it is assigned the name /dev/mmcblk0 and the mounting freezes for 90 seconds before the controller completes startup.

**Bug 1126501: After updating all system apps on devices with image 2.6, "Upgrade ctrlX OS" (remodeling) cannot be executed****Description**

On devices with image 2.6, it is not possible to run "Upgrade ctrlX OS" (remodeling) after updating all system apps. It is canceled with an error message.

### Workaround

The current image with version 4.4 has to be installed. The bug is planned to be fixed in the next version 4.6.

#### Bug 1129085: Image update does not work reliably on ctrlX CORE X5

##### Description

The internally connected eMMC controller is not reliably initialized as mmcblk0 on ctrlX CORE X5 with kernel 6.12 (included in image 4.4), but with a probability of 50% as mmcblk1.

For the image update, the file rescue.toml is written, which defines the target medium for the image.

If the mmcblk1 medium is now detected as the boot medium before the image update, this is also written to rescue.toml and the image update cannot be carried out.

### Workaround

Before executing an image update, check which medium is detected as SystemData. If it is not mmcblk0, reboot the system prior to updating the image.

## 4.4.5 Setup

#### Bug 1116120: Backup/restore may fail after 30 days operating time of the device

##### Description

Backup/restore and requires an app-specific temporary folder. This folder is created upon startup. With CORE24, a change was introduced with regard to cleaning up temporary folders. With the current version used by ctrlX OS, temporary folders that are not used for 30 days are removed. After removing the temporary folders, an app can no longer access its app-specific temporary folder, which can result in various errors, e.g. backup/restore can no longer fulfill its task, apps can no longer be installed, etc.

### Workaround

If the operating time of a device is more than 30 days and apps cannot perform their tasks, it may be helpful to restart the device. The desired task should then work again, provided the error was related to the problem mentioned.

## 4.4.6 Solutions

#### Bug 1117106: Files with non-ASCII characters in the name are restored incorrectly when uploading

##### Description

If a configuration contains files whose file names contain non-ASCII characters and this configuration is archived with a ZIP program that does not automatically use UTF8 for such characters (e.g. 7-Zip), the names of these files are restored incorrectly when uploading. The Windows-integrated packer and the Solutions app use UTF8 when creating archives, so that the problem does not occur.

#### Bug 1122488: Incomplete backup of Git settings

##### Description

The settings for the SSH remote server are not backed up and restored.

## 5 Service and support

Our worldwide service network provides an optimized and efficient support. Our experts provide you with advice and assistance. You can contact us **24/7**.

### Service Germany

Our technology-oriented Competence Center in Lohr, Germany, is responsible for all your service-related queries for electric drive and controls.

Contact the **Service Hotline** and **Service Helpdesk** under:

Phone:     **+49 9352 40 5060**  
Fax:        **+49 9352 18 4941**  
Email:      ↪ [service.svc@boschrexroth.de](mailto:service.svc@boschrexroth.de)  
Internet:   ↪ <http://www.boschrexroth.com>

Additional information on service, repair (e.g. delivery addresses) and training can be found on our internet sites.

### Service worldwide

Outside Germany, please contact your local service office first. For hotline numbers, refer to the sales office addresses on the internet.

### Preparing information

To be able to help you more quickly and efficiently, please have the following information ready:

- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)



## 6 Index

### 1, 2, 3 ...

|                                                                          |    |
|--------------------------------------------------------------------------|----|
| "Apps" and "Licenses" were merged .....                                  | 12 |
| "Reset device data" can fail sporadically ...                            | 37 |
| "Upgrade ctrlX OS" (remodeling) fails sporadically .....                 | 30 |
| "Upgrading ctrlX OS" (remodeling) not possible with snapd 2.68.5.1 ..... | 31 |
| 10-day trial period available for selected apps .....                    | 28 |

### A

|                                                                                                                    |        |
|--------------------------------------------------------------------------------------------------------------------|--------|
| A USB flash drive cannot be mounted again if it is removed without clicking on "remove safely" .....               | 30     |
| About this documentation .....                                                                                     | 9      |
| After updating all system apps on devices with image 2.6, "Upgrade ctrlX OS" (remodeling) cannot be executed ..... | 16, 37 |
| Alarm "Failed to set CPU configuration" is displayed .....                                                         | 30     |
| Authorization "Manage configurations" is insufficient for loading and saving .....                                 | 33     |
| Automation Core                                                                                                    |        |
| New functions 4.4.0 .....                                                                                          | 11, 23 |
| Notes on use and known restrictions 4.4.0 .....                                                                    | 18, 35 |
| Resolved defects and modifications 4.4.0 .....                                                                     | 14, 29 |

### B

|                                                                          |        |
|--------------------------------------------------------------------------|--------|
| Backup/restore does not support NFS and SMB mount points .....           | 32     |
| Backup/restore may fail after 30 days operating time of the device ..... | 15, 38 |
| Balanced energy mode for ctrlX OS (Balanced Mode) .....                  | 27     |
| Bug 589936 .....                                                         | 18     |
| Bug 711581 .....                                                         | 19     |
| Bug 853922 .....                                                         | 19     |
| Bug 855538 .....                                                         | 37     |
| Bug 867074 .....                                                         | 19     |
| Bug 899934 .....                                                         | 20     |
| Bug 940111 .....                                                         | 29     |
| Bug 970580 .....                                                         | 33     |
| Bug 994012 .....                                                         | 30     |
| Bug 995534 .....                                                         | 30     |
| Bug 1010221 .....                                                        | 29     |
| Bug 1014687 .....                                                        | 20     |
| Bug 1015711 .....                                                        | 31     |
| Bug 1021621 .....                                                        | 20, 37 |
| Bug 1035162 .....                                                        | 31     |
| Bug 1035172 .....                                                        | 31     |
| Bug 1041006 .....                                                        | 32     |
| Bug 1043154 .....                                                        | 30     |
| Bug 1049110 .....                                                        | 30     |
| Bug 1051857 .....                                                        | 35     |
| Bug 1053299 .....                                                        | 32     |

|                                                                          |        |
|--------------------------------------------------------------------------|--------|
| Bug 1062953 .....                                                        | 32     |
| Bug 1064879 .....                                                        | 30     |
| Bug 1068029 .....                                                        | 32     |
| Bug 1070473 .....                                                        | 33     |
| Bug 1071563 .....                                                        | 32     |
| Bug 1076361 .....                                                        | 15, 37 |
| Bug 1076429 .....                                                        | 32     |
| Bug 1077634 .....                                                        | 32     |
| Bug 1077648 .....                                                        | 33     |
| Bug 1079332 .....                                                        | 33     |
| Bug 1079925 .....                                                        | 31     |
| Bug 1081521 .....                                                        | 31     |
| Bug 1091116 .....                                                        | 14     |
| Bug 1098779 .....                                                        | 14, 36 |
| Bug 1100451 .....                                                        | 14, 36 |
| Bug 1111432 .....                                                        | 36     |
| Bug 1111609 .....                                                        | 20, 37 |
| Bug 1113522 .....                                                        | 33     |
| Bug 1114792 .....                                                        | 11     |
| Bug 1114982 .....                                                        | 12     |
| Bug 1116120 .....                                                        | 15, 38 |
| Bug 1116146 .....                                                        | 15, 37 |
| Bug 1117106 .....                                                        | 38     |
| Bug 1117971 .....                                                        | 29     |
| Bug 1120682 .....                                                        | 36     |
| Bug 1122488 .....                                                        | 16, 38 |
| Bug 1122688 .....                                                        | 36     |
| Bug 1124644 .....                                                        | 15     |
| Bug 1126017 .....                                                        | 15, 36 |
| Bug 1126501 .....                                                        | 16, 37 |
| Bug 1126921 .....                                                        | 31     |
| Bug 1129085 .....                                                        | 16, 38 |
| Bug 1131271 .....                                                        | 36     |
| Bug 1131434 .....                                                        | 12     |
| Bug 1133074 .....                                                        | 16     |
| Bug 1134469 .....                                                        | 16     |
| Bug 1139268 .....                                                        | 15     |
| Bug 1179239 .....                                                        | 20     |
| Bug 1188123 .....                                                        | 20     |
| Bug 1189755 .....                                                        | 21     |
| Bug 1192160 .....                                                        | 21     |
| Bug 1192174 .....                                                        | 21     |
| Bugfix for sporadic startup problem due to unstable eMMC numbering ..... | 14     |

### C

|                                                               |        |
|---------------------------------------------------------------|--------|
| Certain snapd versions cannot be restored .                   | 32     |
| Certificate and PKI cannot be deleted together .....          | 33     |
| Changes in the Data Layer .....                               | 12, 24 |
| Collaborative engineering with Git .....                      | 29     |
| Configurations can be added to version control with Git ..... | 28     |
| Creating a system report seems to freeze ...                  | 21     |
| Creating a user account during initial commissioning .....    | 13     |

ctrlX CORE X5: Device does sporadically not start if an external data carrier is plugged into port XF01A, XF01B or CF01 . . . . . 19

ctrlX CORE X5/X7: First boot after image update (hardware reset) takes longer . . . . . 20

ctrlX CORE X5/X7: Warning 080E082A/0C64100D "App connection restored" after image update . . . . . 36

ctrlX CORE X7 startup sporadically delayed by 140 seconds . . . . . 36

ctrlX OS sporadically takes longer to boot on ctrlX CORE . . . . . 15, 37

## D

Data Layer improvements . . . . . 25

Deleting backup files after restoring . . . . . 28

Device Admin

- Bugfixes and modifications 4.4.0 . . . . 15, 29
- New functions 4.4.0 . . . . . 12, 27
- Notes on use and known restrictions 4.4.0 . . . . . 20, 37

Device Admin supports CAN interfaces . . . . 27

Device does not start with USB flash drive inserted on XF01C . . . . . 19

DHCP server support in ctrlX OS . . . . . 12

Display CPU information in "System information" . . . . . 28

Display of remaining time in trial mode . . . . 28

Displaying the ctrlX OS version in "System information" . . . . . 28

## E

Error when automatically creating callables . 29

Error when creating the callable during start-up . . . . . 29

Error when restoring a subscription (certificate management) . . . . . 31

Error when restoring PKIs . . . . . 32

Extended Git functionality . . . . . 14

Extended time synchronization (e.g. PTP) resulted in jitter in real-time processes . . . . 15

## F

Feature 282405 . . . . . 12

Feature 548599 . . . . . 27

Feature 740428 . . . . . 23

Feature 836887 . . . . . 27

Feature 870976 . . . . . 23

Feature 914768 . . . . . 27

Feature 938847 . . . . . 24

Feature 948328 . . . . . 27

Feature 981999 . . . . . 28

Feature 984227 . . . . . 24

Feature 986338 . . . . . 27

Feature 986818 . . . . . 27

Feature 995017 . . . . . 24

Feature 995857 . . . . . 28

Feature 995988 . . . . . 29

Feature 995994 . . . . . 14

Feature 996130 . . . . . 24

Feature 996303 . . . . . 27

Feature 1006016 . . . . . 27

Feature 1032629 . . . . . 27

Feature 1040678 . . . . . 28

Feature 1041322 . . . . . 12

Feature 1045881 . . . . . 25

Feature 1047945 . . . . . 26

Feature 1048023 . . . . . 26

Feature 1048209 . . . . . 28

Feature 1051992 . . . . . 13

Feature 1054708 . . . . . 28

Feature 1058176 . . . . . 13

Feature 1059916 . . . . . 26

Feature 1060885 . . . . . 28

Feature 1061268 . . . . . 28

Feature 1071645 . . . . . 28

Feature 1112318 . . . . . 13

Feature 1112320 . . . . . 13

Feature 1155681 . . . . . 13

Files with non-ASCII characters in the name are restored incorrectly when uploading . . . . . 38

Fixed bugs 3.6.1 . . . . . 14, 29

Flexible core division . . . . . 26

Frequent error 081F0042/0C7B4285 when switching from OPERATING to SERVICE . 15, 36

Functional extensions in the Data Layer web interface . . . . . 26

Functional extensions in the logbook web interface . . . . . 23

## H

Hardware support cannot be updated on a control with Secure Boot (ctrlX CORE X5/X7) from the store . . . . . 20, 37

Helpdesk . . . . . 39

Hotline . . . . . 39

## I

Image update does not work reliably on ctrlX CORE X5 . . . . . 16, 38

Important information . . . . . 9

Improved display of the login banner for ctrlX OS . . . . . 28

Improved handling of Motion errors . . . . . 23

Incomplete backup of Git settings . . . . . 16, 38

Increased runtime of scheduler tasks . . . . . 36

Increasing "deviation" time measurement for scheduler tasks . . . . . 36

Information about our products . . . . . 9

Interrupting real-time tasks at full core load . 20

## K

Known restrictions 4.4.0 . . . . . 16, 33

## L

License server is not enabled when restoring a backup . . . . . 33

Linux Kernel Update . . . . . 12

|                                                                                                 |        |                                                                                                                                      |        |
|-------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------|--------|
| Linux kernel updated to version 6.12.40 . . . .                                                 | 26     | Supporting VLAN via DAN network interfaces                                                                                           | 27     |
| Logbook missing from system report . . . . .                                                    | 16     | System image update from version 2.6 and older to version 4.4 via the ctrlX OS web interface results in an error . . . . .           | 14, 36 |
| <b>M</b>                                                                                        |        | System message "Cannot find unit for notify message of PID ****, ignoring." in the logbook . . . . .                                 | 14, 36 |
| Memory leak in UPNP service . . . . .                                                           | 16     | System report . . . . .                                                                                                              | 27     |
| Mounting and unmounting storage media using Data Layer commands . . . . .                       | 27     | <b>T</b>                                                                                                                             |        |
| <b>N</b>                                                                                        |        | The system operating time is not updated in the Data Layer . . . . .                                                                 | 31     |
| New functions 4.4.0 . . . . .                                                                   | 23     | Time and time zone are not applied . . . .                                                                                           | 15, 37 |
| New functions 4.6.0 . . . . .                                                                   | 11     | <b>U</b>                                                                                                                             |        |
| Notification for differing time . . . . .                                                       | 27     | Unauthenticated access to API endpoint . . .                                                                                         | 31     |
| <b>P</b>                                                                                        |        | Unauthorized access to sensitive data via security mechanisms . . . . .                                                              | 31     |
| Packet loss during redundant communication with HSR . . . . .                                   | 19     | Unencrypted backup files . . . . .                                                                                                   | 32     |
| Pending error after image update . . . . .                                                      | 35     | Uninstall button for the system app "core24" after remodeling . . . . .                                                              | 21     |
| Pending information messages in alarms . . .                                                    | 11     | Uninstallation takes a very long time . . . .                                                                                        | 20     |
| Power controller firmware is not updated . .                                                    | 15     | Updating all apps on a device with image version 2.6.10 aborts sporadically after the core app . . . . .                             | 20     |
| Public keys can be copied to the clipboard .                                                    | 13     | Updating all apps on a device with image version 2.6.10 is canceled after the Core22 app . . . . .                                   | 37     |
| <b>R</b>                                                                                        |        | Updating the "core" system app . . . . .                                                                                             | 24     |
| Renaming "Firmware update" to "Flash system image" . . . . .                                    | 27     | Updating the apps on ctrlX CORE 4.4 . . .                                                                                            | 17, 34 |
| Resetting the default password is not possible . . . . .                                        | 31     | Updating the apps on the ctrlX CORE Virtual 4.4.0 . . . . .                                                                          | 16, 33 |
| Resetting the device data fails . . . . .                                                       | 20     | Updating the Celix Framework . . . . .                                                                                               | 24     |
| Restoring a backup on a control of a different type fails . . . . .                             | 21     | Uploading archives with read-only directories fails . . . . .                                                                        | 33     |
| <b>S</b>                                                                                        |        | User is not removed from group . . . . .                                                                                             | 32     |
| SD card cannot be read in Windows PC sporadically after firmware update . . . . .               | 29     | User-defined password policy is not deleted when restoring a backup . . . . .                                                        | 32     |
| Server-based licensing is not enabled when restoring a backup . . . . .                         | 32     | <b>V</b>                                                                                                                             |        |
| Service hotline . . . . .                                                                       | 39     | Various functional adjustments to the Data Layer web interface . . . . .                                                             | 24     |
| Setup                                                                                           |        | <b>W</b>                                                                                                                             |        |
| New functions 4.4.0 . . . . .                                                                   | 28     | Warning "Json schema issue: ..." when restoring a backup . . . . .                                                                   | 33     |
| New functions 4.6.0 . . . . .                                                                   | 13     | Warning of downgrade . . . . .                                                                                                       | 13     |
| Notes on the use and known restrictions .                                                       | 21     | When formatting a partition, the name of the storage medium must be specified . . . . .                                              | 30     |
| Notes on use and known restrictions . . .                                                       | 38     | <b>X</b>                                                                                                                             |        |
| Resolved defects and modifications 4.4.0                                                        | 31     | X3: Retain memory > 56 kB sporadically inconsistent after switching off/on in case of a drive-integrated control and ctrlX CORE X3 . | 18     |
| Simple Network Management Protocol can be used in ctrlX OS . . . . .                            | 27     |                                                                                                                                      |        |
| Snapd app update fails for devices without serial assertion . . . . .                           | 30     |                                                                                                                                      |        |
| Software and apps 4.4.0 . . . . .                                                               | 21     |                                                                                                                                      |        |
| Software and apps 4.6.0 . . . . .                                                               | 9      |                                                                                                                                      |        |
| Solutions                                                                                       |        |                                                                                                                                      |        |
| New functions 4.4.0 . . . . .                                                                   | 28     |                                                                                                                                      |        |
| New functions 4.6.0 . . . . .                                                                   | 14     |                                                                                                                                      |        |
| Notes on the use and known restrictions .                                                       | 38     |                                                                                                                                      |        |
| Resolved defects and modifications 4.4.0                                                        | 16, 33 |                                                                                                                                      |        |
| Support . . . . .                                                                               | 39     |                                                                                                                                      |        |
| Support for embedding third-party applications in the ctrlX user interface via iFrame . . . . . | 13     |                                                                                                                                      |        |
| Support of further settings . . . . .                                                           | 13     |                                                                                                                                      |        |
| Supporting additional settings . . . . .                                                        | 28     |                                                                                                                                      |        |
| Supporting Single Sign-on (SSO) . . . . .                                                       | 27     |                                                                                                                                      |        |





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